CharDB
Final Assignment

Data Warehouse Design & Implementation

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Executive Summary

This assignment presents the design and implementation of a **star schema data warehouse** optimized for analyzing sales funnel performance. The CharDB model enables stakeholders to track lead progression across multiple sales phases, identify conversion patterns, and derive actionable insights through efficient analytical queries. The implementation includes a robust data cleaning pipeline and strategic indexing to ensure data quality and optimal query performance.

1. Data Model Description

The data model follows a **star schema architecture**, specifically designed for business intelligence and analytical processing use cases. Its primary objective is to support comprehensive analysis of the sales funnel, identify key factors influencing conversion rates, and pinpoint drop-off patterns across the customer journey.

1.1 Core Architecture

At the center of this architecture is the **fact table fct_sales_phases**, which contains one record per lead. This table captures the principal business events and metrics related to each lead throughout the entire sales funnel.

Surrounding the fact table are several **dimension tables** that provide rich descriptive context:

- **dim_zipcode:** Supplies geographical information including province, autonomous community, and geographical coordinates
- **dim_date:** Enables temporal analysis by decomposing dates into day, month, quarter, and year hierarchies
- **dim_financing_type:** Describes the financing option associated with each lead
- **dim_phase:** Represents both the current sales phase and pre-KO phase status of each lead

This structure allows analysts and business users to slice and aggregate data efficiently across multiple dimensions such as time, location, financing type, and sales phase.

2. Design Rationale

The decision to implement a **star schema** was driven by four critical business and technical requirements:

2.1 Simplicity

The model is intuitively designed for both technical and business users, with clear separation of concerns:

- **Facts (measurable events)** are stored in fct_sales_phases
- **Dimensions (descriptive attributes)** are stored in dedicated dimension tables

This clarity makes the model accessible across organizational levels and reduces the learning curve for new users.

2.2 Performance Optimization

Star schemas are highly efficient for analytical workloads:

- **Joins are simplified** (fact table connects directly to dimensions)
- **Data is strategically denormalized** to reduce computational overhead
- **Aggregations and KPI calculations execute faster**

This architecture is particularly efficient for computing critical business metrics: conversion rate, KO (knock-out) rate, dismissal rate, and average installation price.

2.3 Scalability

The model's design inherently supports growth and evolution:

- **New dimensions** can be easily added (e.g., customer segmentation, product categories)
- **Additional measures** can be incorporated into the fact table
- **New data sources** can be integrated without restructuring the core model

This flexibility ensures the model remains relevant as business requirements and organizational priorities evolve.

2.4 Analytical Capability

The structure enables sophisticated multi-dimensional analysis:

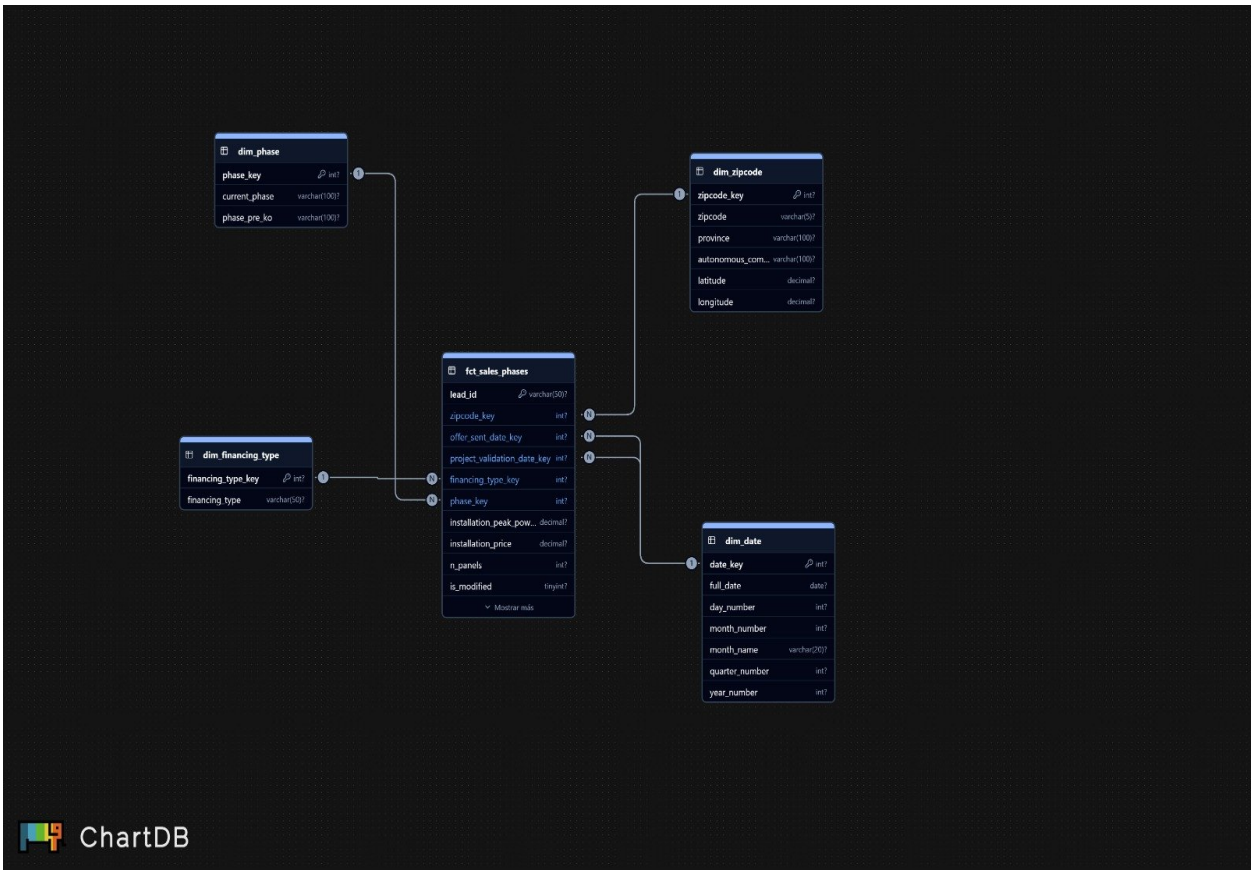
- **Temporal analysis:** Funnel performance trends over time using dim_date
- **Geographic analysis:** Performance variation across regions using dim_zipcode
- **Financing analysis:** Conversion patterns by financing type using dim_financing_type
- **Phase-based analysis:** Drop-off identification by sales phase using dim_phase

3. Schema Illustration

The star schema follows the classic hub-and-spoke topology:

- **The fact table fct_sales_phases** is positioned at the center
- **Dimension tables** are connected through foreign key relationships
- **All relationships** are one-to-many from dimensions to the fact table

The diagram illustrates the complete schema structure and relationships, demonstrating how the model is optimized for analytical queries and multi-dimensional analysis.



4. Data Cleaning Operations

Data quality is foundational to reliable analytics. The project implements a comprehensive data cleaning pipeline in the **staging layer** using dbt (data build tool), ensuring data consistency, accuracy, and reliability before loading into the analytical model.

4.1 Text Standardization

Fields processed: zipcode, province, city, region, financing_type
Operations: TRIM() removes leading and trailing whitespace; UPPER() enforces consistent capitalization
Benefits: Eliminates variations caused by inconsistent data entry, prevents duplicate records, and ensures reliable joins and aggregations.

4.2 Zipcode Standardization

Operation: LPAD(TRIM(zipcode), 5, '0')
Benefits: Ensures all zipcodes conform to a standard 5-digit format, prevents join failures due to mismatched formats, and improves data integrity across geographic analyses.

4.3 Null and Empty Value Handling

Operation: NULLIF(TRIM(column), '')
Benefits: Distinguishes between genuinely missing data and empty strings, prevents incorrect values from distorting aggregations, and improves accuracy of analytical queries.

4.4 Data Type Casting

Operations: Dates → DATE; Numeric values → DECIMAL; Boolean flags → TINYINT
Examples: CAST(installation_price AS DECIMAL(12,2)) and STR_TO_DATE(date, '%Y-%m-%d')
Benefits: Ensures correct mathematical operations, prevents runtime errors, and improves query execution speed and efficiency.

4.5 Additional Cleaning Operations

Invalid Record Filtering: Records missing critical fields are excluded to ensure only validated, complete records are used in analysis.
Deduplication: SELECT DISTINCT eliminates double-counting and ensures accuracy of all KPI calculations.
Derived Columns: New analytical flags (is_converted, is_ko, is_dismissed) are created to simplify downstream queries and improve maintainability.

5. Indexing Strategy in the Data Warehouse

Indexing is critical for maintaining query performance as data volumes grow. The warehouse implements a strategic indexing approach targeting the most frequently accessed columns and join patterns.

5.1 Primary Key Indexes

Automatically created on unique identifiers across all tables (lead_id, zipcode_key, date_key, financing_type_key, phase_key).
Benefits: Enforces uniqueness constraints, dramatically improves lookup performance, and is essential for join operations.

5.2 Foreign Key Indexes

Applied on all foreign keys in the fact table to optimize dimension table joins (zipcode_key, offer_sent_date_key, project_validation_date_key, financing_type_key, phase_key).
Benefits: Accelerates fact-to-dimension joins, reduces overall query execution time, and is essential for star join optimization.

5.3 Temporal and Filter Indexes

Specialized indexes on date-related columns (offer_sent_date, project_validation_date) and frequently used filters (financing_type, current_phase).
Benefits: Accelerates time-based filtering, improves KPI calculations, enables fast segmentation analysis, and improves dashboard load times.

5.4 Composite Indexes

Multi-column indexes (e.g., zipcode_key, offer_sent_date_key) optimize queries with multiple filter conditions.
Benefits: Optimizes queries filtering by location and time simultaneously, improves performance of grouped aggregations, and reduces need for multiple index scans.

6. Conclusion

The CharDB data warehouse represents a comprehensive, well-architected solution for sales funnel analytics. By combining a clean, logical star schema design with rigorous data quality processes and strategic indexing, the model delivers exceptional value:

1. **Accessibility:** Intuitive schema design enables business users to self-serve analytics
2. **Performance:** Optimized queries return results in seconds, supporting real-time dashboards
3. **Reliability:** Comprehensive cleaning ensures accurate, trustworthy insights
4. **Scalability:** Flexible architecture supports future expansion and new analytical requirements
5. **Maintainability:** Clear structure and documentation reduce operational overhead

This foundation positions the organization to extract maximum business value from lead and sales data, enabling data-driven decision-making across the sales organization and beyond.

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